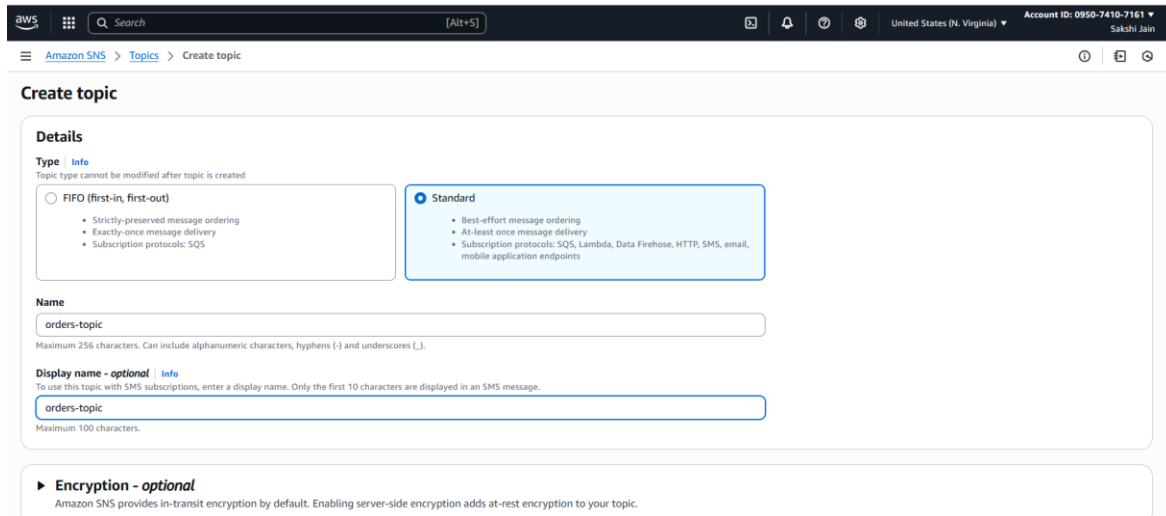


Serverless Order Notification & Processing System

1. Create SNS topic

Console: Amazon SNS → Topics → Create topic

- Type: Standard
- Name: orders-topic
- Leave defaults



The screenshot shows the 'Create topic' page in the AWS Management Console. The breadcrumb trail is 'Amazon SNS > Topics > Create topic'. The page title is 'Create topic'. Under the 'Details' section, the 'Type' is set to 'Standard' (selected with a radio button). The 'Name' field contains 'orders-topic'. The 'Display name - optional' field also contains 'orders-topic'. A note indicates that the topic type cannot be modified after creation. A section for 'Encryption - optional' is visible at the bottom, stating that Amazon SNS provides in-transit encryption by default.

2. Create SQS queues (3) + DLQs

Create three main queues and a DLQ for each (or shared DLQ).

Console: Amazon SQS → Create queue

- Payment queue: payment-queue (Standard)
- Inventory queue: inventory-queue
- Notification queue: notification-queue

Amazon SQS > Queues > Create queue

Create queue

Details

Type

Choose the queue type for your application or cloud infrastructure.

Standard Info

At-least-once delivery, message ordering isn't preserved

At-least once delivery

Best-effort ordering

FIFO Info

First-in-first-out delivery, message ordering is preserved

First-in-first-out delivery

Exactly-once processing

You can't change the queue type after you create a queue.

Name

payment-queue

A queue name is case-sensitive and can have up to 80 characters. You can use alphanumeric characters, hyphens (-), and underscores (_).

Configuration Info

Set the maximum message size, visibility to other consumers, and message retention.

Visibility timeout Info

30

Seconds

Message retention period Info

4

Days

Amazon SQS > Queues > Create queue

Create queue

Details

Type

Choose the queue type for your application or cloud infrastructure.

Standard Info

At-least-once delivery, message ordering isn't preserved

At-least once delivery

Best-effort ordering

FIFO Info

First-in-first-out delivery, message ordering is preserved

First-in-first-out delivery

Exactly-once processing

You can't change the queue type after you create a queue.

Name

inventory-queue

A queue name is case-sensitive and can have up to 80 characters. You can use alphanumeric characters, hyphens (-), and underscores (_).

Configuration Info

Set the maximum message size, visibility to other consumers, and message retention.

Visibility timeout Info

30

Seconds

Message retention period Info

4

Days

aws

Q Search

[Alt+S]

United States (N. Virginia)

Account ID: 0950-7410-7161

Sakshi Jain

Amazon SQS > Queues > Create queue

Create queue

Details

Type

Choose the queue type for your application or cloud infrastructure.

Standard Info

At-least-once delivery, message ordering isn't preserved

At-least once delivery

Best-effort ordering

FIFO Info

First-in-first-out delivery, message ordering is preserved

First-in-first-out delivery

Exactly-once processing

You can't change the queue type after you create a queue.

Name

notification-queue

A queue name is case-sensitive and can have up to 80 characters. You can use alphanumeric characters, hyphens (-), and underscores (_).

Configuration Info

Set the maximum message size, visibility to other consumers, and message retention.

Visibility timeout Info

30

Seconds

Message retention period Info

4

Days

Amazon SQS > Queues

Queues (3)

Q Search queues by prefix

Create

Edit

Delete

Send and receive messages

Actions

Create queue

< 1 >

	Name	Type	Created	Messages available	Messages in flight	Encryption	Content-based deduplication
☐	inventory-queue	Standard	2025-12-11T10:31+05:30	0	0	Amazon SQS key (SSE-SQS)	-
☐	notification-queue	Standard	2025-12-11T10:32+05:30	0	0	Amazon SQS key (SSE-SQS)	-
☐	payment-queue	Standard	2025-12-11T10:30+05:30	0	0	Amazon SQS key (SSE-SQS)	-

3. Subscribe SQS queues to SNS topic

Console: SNS → Topic → Subscriptions → Create subscription

- Protocol: SQS
- Endpoint: pick SQS queue ARN

aws Search [Alt+S] United States (N. Virginia) Account ID: 0950-7410-7161 Sakshi Jain

Amazon SNS Subscriptions Create subscription

Create subscription

Details

Topic ARN
arn:aws:sns:us-east-1:095074107161:orders-topic

Protocol
The type of endpoint to subscribe
Amazon SQS

Endpoint
Only Amazon SQS standard queues will be listed and can receive notifications from an Amazon SNS standard topic.
arn:aws:sqs:us-east-1:095074107161:inventory-queue

☐ Enable raw message delivery

After your subscription is created, you must confirm it. [Info](#)

► **Subscription filter policy - optional** [Info](#)
This policy filters the messages that a subscriber receives.

Amazon SNS Subscriptions Create subscription

Create subscription

Details

Topic ARN
arn:aws:sns:us-east-1:095074107161:orders-topic

Protocol
The type of endpoint to subscribe
Amazon SQS

Endpoint
Only Amazon SQS standard queues will be listed and can receive notifications from an Amazon SNS standard topic.
arn:aws:sqs:us-east-1:095074107161:notification-queue

☐ Enable raw message delivery

After your subscription is created, you must confirm it. [Info](#)

► **Subscription filter policy - optional** [Info](#)
This policy filters the messages that a subscriber receives.

Amazon SNS Subscriptions Create subscription

Create subscription

Details

Topic ARN
arn:aws:sns:us-east-1:095074107161:orders-topic

Protocol
The type of endpoint to subscribe
Amazon SQS

Endpoint
Only Amazon SQS standard queues will be listed and can receive notifications from an Amazon SNS standard topic.
arn:aws:sqs:us-east-1:095074107161:payment-queue

☐ Enable raw message delivery

After your subscription is created, you must confirm it. [Info](#)

► **Subscription filter policy - optional** [Info](#)
This policy filters the messages that a subscriber receives.

Subscriptions (3)

[Edit](#)[Delete](#)[Request confirmation](#)[Confirm subscription](#)[Create subscription](#)

ID	Endpoint	Status	Protocol	Topic
0146a907-8ea3-4d5a-902e-2...	arn:aws:sqs:us-east-1:095074...	Confirmed	SQS	orders-topic
7cc8195a-5d58-44dc-bf19-f8...	arn:aws:sqs:us-east-1:095074...	Confirmed	SQS	orders-topic
83e88274-3c2a-4806-85a9-0...	arn:aws:sqs:us-east-1:095074...	Confirmed	SQS	orders-topic

4. Create IAM role(s) for Lambda functions

Each Lambda needs an execution role with:

- AWSLambdaBasicExecutionRole (CloudWatch logs)
- Permissions for SQS (Lambda will read via event mapping, but for uploads/reads to other services like DB/SES you'll add)
- If lambdas access DynamoDB or RDS or SNS/SES, add those permissions.

[IAM](#) > [Roles](#) > [PaymentLambda-role-nk1zuw8p](#) > Add Permissions

Attach policy to PaymentLambda-role-nk1zuw8p

Current permissions policies (1)

Other permissions policies (1/1107)

Policy name	Type	Description
☒ AWSLambdaSQSQueueExecutionRole	AWS managed	Provides receive message, delete messag...

[Cancel](#)[Add permissions](#)

[IAM](#) > [Roles](#)

Identity and Access Management (IAM)

[Search IAM](#)

Dashboard

Access management

- User groups
 - Users
 - Roles**
 - Policies
 - Identity providers
 - Account settings
 - Root access management
 - Temporary delegation requests
 - [New](#)
- Access reports**
 - Access Analyzer
 - Resource analysis [New](#)
 - Unused access
 - [Analyze settings](#)

Roles (1/10) Info

An IAM role is an identity you can create that has specific permissions with credentials that are valid for short durations. Roles can be assumed by entities that you trust.

Role name	Trusted entities	Last activity
☒ InventoryLambda-role-llvrz5f1	AWS Service: lambda	-

Roles Anywhere Info

Authenticate your non AWS workloads and securely provide access to AWS services.



Access AWS from your non AWS workloads

Operate your non AWS workloads using the same authentication and authorization strategy that you use within AWS.



X.509 Standard

Use your own existing PKI infrastructure or use [AWS Certificate Manager Private Certificate Authority](#) to authenticate identities.



Temporary credentials

Use temporary credentials with ease and benefit from the enhanced security they provide.

[Manage](#)

[CloudShell](#) [Feedback](#) [Console Mobile App](#)

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IAM > Roles > InventoryLambda-role-llvrz5f1 > Add Permissions

Attach policy to InventoryLambda-role-llvrz5f1

► Current permissions policies (1)

Other permissions policies (1/1107)

Filter by Type

Q awsiamdasq X All types 1 match

☒	Policy name	Type	Description
☒	AWSLambdaSQSQueueExecutionRole	AWS managed	Provides receive message, delete messag...

Cancel Add permissions

IAM > Roles > NotificationLambda-role-s1kbayzf

NotificationLambda-role-s1kbayzf Info

Summary

Creation date
December 11, 2025, 10:51 (UTC+05:30)

ARN
 arn:aws:iam:095074107161:role/service-role/NotificationLambda-role-s1kbayzf

Last activity
-

Maximum session duration
1 hour

Delete Edit

Permissions Trust relationships Tags Last Accessed Revoke sessions

Permissions policies (1) Info

You can attach up to 10 managed policies.

Filter by Type

Q Search All types 1

☐	Policy name	Type	Attached entities
☐	AWSLambdaBasicExecutionRole-Sd5a3775-73...	Customer managed	1

Simulate Remove Add permissions

► Permissions boundary (not set)

IAM > Roles > NotificationLambda-role-s1kbayzf > Add Permissions

Attach policy to NotificationLambda-role-s1kbayzf

► Current permissions policies (1)

Other permissions policies (1/1107)

Filter by Type

Q awsiamdasq X All types 1 match

☒	Policy name	Type	Description
☒	AWSLambdaSQSQueueExecutionRole	AWS managed	Provides receive message, delete messag...

Cancel Add permissions

5. Create three Lambda functions

Create PaymentLambda, InventoryLambda, NotificationLambda.

Settings:

- Runtime: Python 3.12 (or NodeJS)
- Timeout: 30–120s depending on work
- Memory: 256–1024 MB depending on workload

PaymentLambda

Lambda > Functions > Create function

Create function [Info](#)

Choose one of the following options to create your function.

☒ Author from scratch

Start with a simple Hello World example.

☐ Use a blueprint

Build a Lambda application from sample code and configuration presets for common use cases.

☐ Container image

Select a container image to deploy for your function.

Basic information

Function name

Enter a name that describes the purpose of your function.

PaymentLambda

Function name must be 1 to 64 characters, must be unique to the Region, and can't include spaces. Valid characters are a-z, A-Z, 0-9, hyphens (-), and underscores (_).

Runtime [Info](#)

Choose the language to use to write your function. Note that the console code editor supports only Node.js, Python, and Ruby.

Python 3.12

Architecture [Info](#)

Choose the instruction set architecture you want for your function code.

☐ arm64

☒ x86_64

Permissions [Info](#)

By default, Lambda will create an execution role with permissions to upload logs to Amazon CloudWatch Logs. You can customize this default role later when adding triggers.

Lambda > Functions > Create function

☐ arm64

☒ x86_64

Permissions [Info](#)

By default, Lambda will create an execution role with permissions to upload logs to Amazon CloudWatch Logs. You can customize this default role later when adding triggers.

▼ Change default execution role

Execution role

Choose a role that defines the permissions of your function. To create a custom role, go to the [IAM console](#).

☐ Create a new role with basic Lambda permissions

☒ Use an existing role

☐ Create a new role from AWS policy templates

Existing role

Choose an existing role that you've created to be used with this Lambda function. The role must have permission to upload logs to Amazon CloudWatch Logs.

service-role/PaymentLambda-role-nk1zuw8p

[View the PaymentLambda-role-nk1zuw8p role](#) on the IAM console.

► Additional configurations

Use additional configurations to set up networking, security, and governance for your function. These settings help secure and customize your Lambda function deployment.

Cancel

Create function

InventoryLambda

✔ Successfully created the function **PaymentLambda**. You can now change its code and configuration. To invoke your function with a test event, choose "Test".

Create function [Info](#)

Choose one of the following options to create your function.

☒ **Author from scratch**

Start with a simple Hello World example.

☐ **Use a blueprint**

Build a Lambda application from sample code and configuration presets for common use cases.

☐ **Container image**

Select a container image to deploy for your function.

Basic information

Function name

Enter a name that describes the purpose of your function.

InventoryLambda

Function name must be 1 to 64 characters, must be unique to the Region, and can't include spaces. Valid characters are a-z, A-Z, 0-9, hyphens (-), and underscores (_).

Runtime [Info](#)

Choose the language to use to write your function. Note that the console code editor supports only Node.js, Python, and Ruby.

Python 3.12

Architecture [Info](#)

Choose the instruction set architecture you want for your function code.

☐ arm64

☒ x86_64

▼ Change default execution role

Execution role

Choose a role that defines the permissions of your function. To create a custom role, go to the [IAM console](#).

☐ Create a new role with basic Lambda permissions

☒ Use an existing role

☐ Create a new role from AWS policy templates

Existing role

Choose an existing role that you've created to be used with this Lambda function. The role must have permission to upload logs to Amazon CloudWatch Logs.

service-role/InventoryLambda-role-llvrz5f1

[View the InventoryLambda-role-llvrz5f1 role](#) on the IAM console.

► Additional configurations

Use additional configurations to set up networking, security, and governance for your function. These settings help secure and customize your Lambda function deployment.

[Cancel](#)

[Create function](#)

NotificationLambda

[Lambda](#)
>
[Functions](#)
>
Create function

Create function

Info

Choose one of the following options to create your function.

☒
Author from scratch
Start with a simple Hello World example.

☐
Use a blueprint
Build a Lambda application from sample code and configuration presets for common use cases.

☐
Container image
Select a container image to deploy for your function.

Basic information

Function name
Enter a name that describes the purpose of your function.

Function name must be 1 to 64 characters, must be unique to the Region, and can't include spaces. Valid characters are a-z, A-Z, 0-9, hyphens (-), and underscores (_).

Runtime | Info
Choose the language to use to write your function. Note that the console code editor supports only Node.js, Python, and Ruby.

Architecture | Info
Choose the instruction set architecture you want for your function code.
☐ arm64
☒ x86_64

Permissions | Info
By default, Lambda will create an execution role with permissions to upload logs to Amazon CloudWatch Logs. You can customize this default role later when adding triggers.

▼ Change default execution role

Execution role
Choose a role that defines the permissions of your function. To create a custom role, go to the [IAM console](#).

☐ Create a new role with basic Lambda permissions
☒ Use an existing role
☐ Create a new role from AWS policy templates

Existing role
Choose an existing role that you've created to be used with this Lambda function. The role must have permission to upload logs to Amazon CloudWatch Logs.

[View the NotificationLambda-role-s1kbayzf role](#) on the IAM console.

► Additional configurations

Use additional configurations to set up networking, security, and governance for your function. These settings help secure and customize your Lambda function deployment.

Cancel

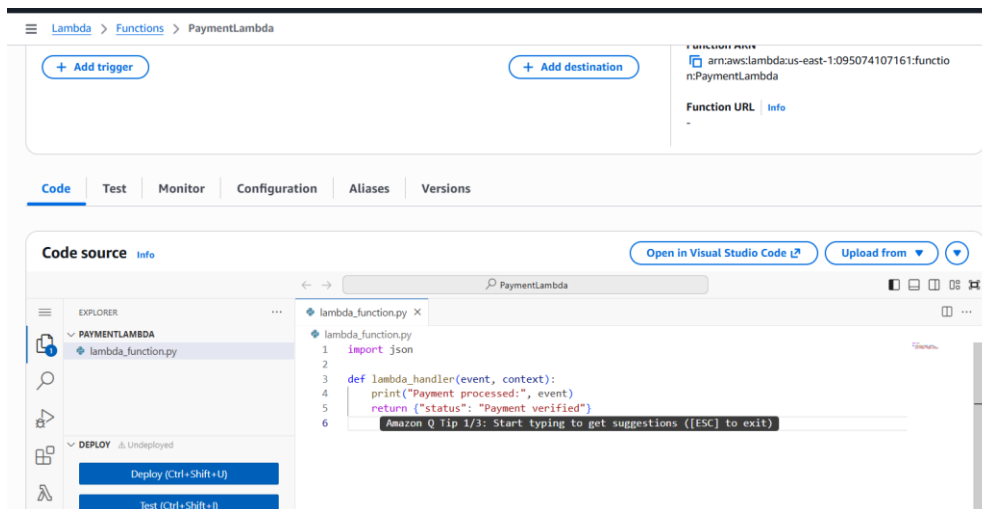
Create function

9. Lambda handler code examples (Python)

PaymentLambda (sample)

```
import json

def lambda_handler(event, context):
    print("Payment processed:", event)
    return {"status": "Payment verified"}
```

Click **Deploy**

InventoryLambda (sample)

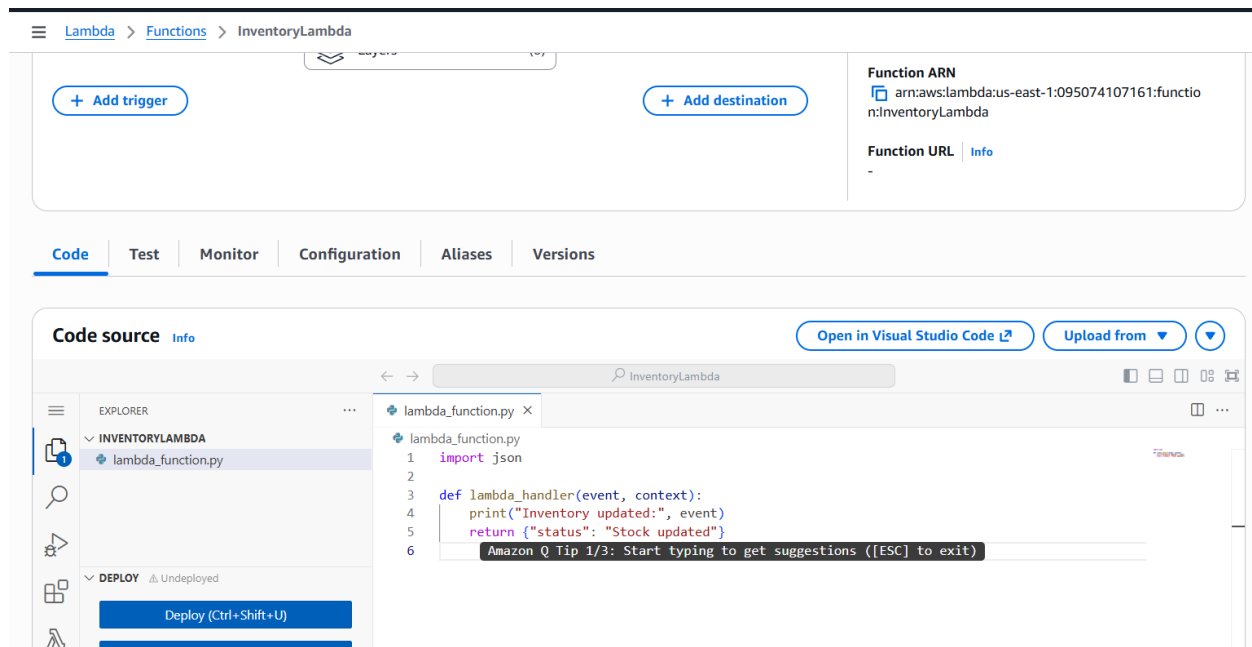
Name: InventoryLambda

Use this code:

```
import json

def lambda_handler(event, context):
    print("Inventory updated:", event)
    return {"status": "Stock updated"}
```

Deploy.



Repeat for NotificationLambda

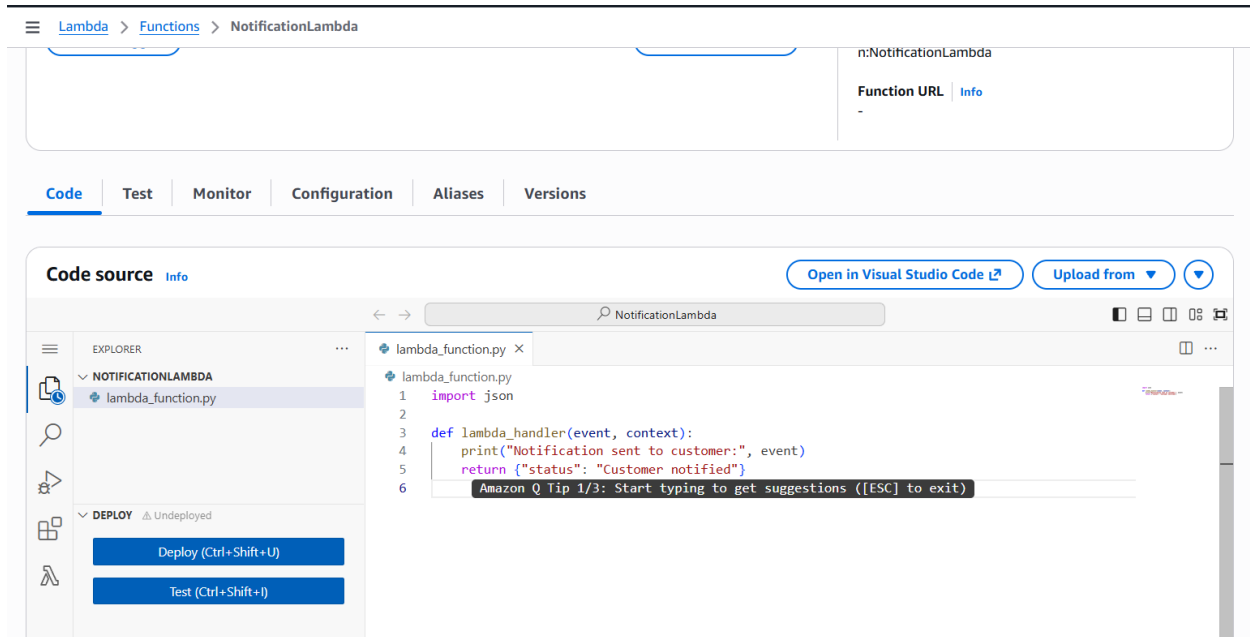
Name: NotificationLambda

Use this code:

```
import json

def lambda_handler(event, context):
    print("Notification sent to customer:", event)
    return {"status": "Customer notified"}
```

Deploy.



10. Connect SQS → Lambda (Triggers)

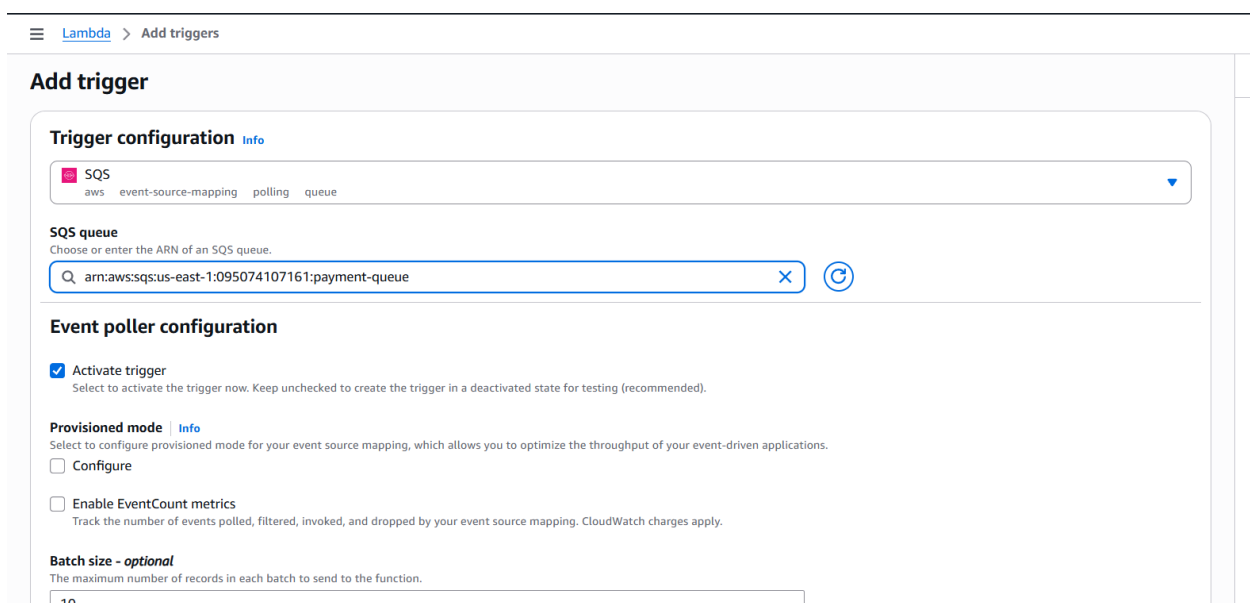
Open PaymentLambda

Scroll → Add Trigger

Choose: **SQS**

Select: PaymentQueue

Click **Add**



Open InventoryLambda

Add trigger → SQS → InventoryQueue

Lambda > Add triggers

Add trigger

Trigger configuration [Info](#)

SQS

aws event-source-mapping polling queue

SQS queue

Choose or enter the ARN of an SQS queue.

Q

arn:aws:sqs:us-east-1:095074107161:inventory-queue

X

↺

Event poller configuration

☒ Activate trigger

Select to activate the trigger now. Keep unchecked to create the trigger in a deactivated state for testing (recommended).

Provisioned mode [Info](#)

Select to configure provisioned mode for your event source mapping, which allows you to optimize the throughput of your event-driven applications.

☐ Configure

☐ Enable EventCount metrics

Track the number of events polled, filtered, invoked, and dropped by your event source mapping. CloudWatch charges apply.

Batch size - optional

The maximum number of records in each batch to send to the function.

10

Open NotificationLambda

Add trigger → SQS → NotificationQueue

Lambda > Add triggers

Add trigger

Trigger configuration [Info](#)

SQS

aws event-source-mapping polling queue

SQS queue

Choose or enter the ARN of an SQS queue.

Q

arn:aws:sqs:us-east-1:095074107161:notification-queue

X

↺

Event poller configuration

☒ Activate trigger

Select to activate the trigger now. Keep unchecked to create the trigger in a deactivated state for testing (recommended).

Provisioned mode [Info](#)

Select to configure provisioned mode for your event source mapping, which allows you to optimize the throughput of your event-driven applications.

☐ Configure

☐ Enable EventCount metrics

Track the number of events polled, filtered, invoked, and dropped by your event source mapping. CloudWatch charges apply.

Batch size - optional

The maximum number of records in each batch to send to the function.

10

11. TEST The Entire Project

Open SNS → OrderTopic

Click: **Publish message**

Step 21 — Enter sample order

Subject: New Order

Message:

```
{
  "orderId": "1001",
  "user": "sakshi",
  "product": "Laptop",
  "price": 55000
}
```

Click **Publish**

Amazon SNS > Topics > orders-topic > Publish message

TIME TO LIVE (TTL) - 300 seconds

This setting applies only to mobile application endpoints. The number of seconds that the push notification service has to deliver the message to the endpoint.

Message body

Message structure

☒ Identical payload for all delivery protocols.
The same payload is sent to endpoints subscribed to the topic, regardless of their delivery protocol.

☐ Custom payload for each delivery protocol.
Different payloads are sent to endpoints subscribed to the topic, based on their delivery protocol.

Message body to send to the endpoint

```
1 {
2   "orderId": "1001",
3   "user": "sakshi",
4   "product": "Laptop",
5   "price": 55000
6 }
7
```

12. Check Each SQS Queue

Messages should be **0 or decreasing**
Means Lambda consumed the message.

Amazon SQS > Queues

Queues (3)

Search queues by prefix

< 1 >

	Name	Type	Created	Messages available	Messages in flight	Encryption	Content-based dedup
☐	inventory-queue	Standard	2025-12-11T10:31+05:30	0	0	Amazon SQS key (SSE-SQS)	-
☐	notification-queue	Standard	2025-12-11T10:32+05:30	0	0	Amazon SQS key (SSE-SQS)	-
☐	payment-queue	Standard	2025-12-11T10:30+05:30	0	0	Amazon SQS key (SSE-SQS)	-

13. Check Logs in CloudWatch

View Logs

Open: CloudWatch → Logs → Log groups

CloudWatch

Favorites and recents

Dashboards

Alarms

AI Operations

GenAI Observability

Application Signals (APM)

Infrastructure Monitoring

Logs

Log Management

Log groups (3)

By default, we only load up to 10,000 log groups.

Filter log groups or try pattern search

Exact match

Log group	Log class	Anomaly d...	Deletion ...	Data pr...	Sensiti...	Retention	Metric ...
/aws/lambda/PaymentLambda	Standard	Configure	Off	-	-	Never exp...	-
/aws/lambda/InventoryLambda	Standard	Configure	Off	-	-	Never exp...	-
/aws/lambda/NotificationLambda	Standard	Configure	Off	-	-	Never exp...	-

CloudWatch

Favorites and recents

Dashboards

Alarms

AI Operations

GenAI Observability

Application Signals (APM)

Infrastructure Monitoring

Logs

Log Management

/aws/lambda/PaymentLambda

Log class: Standard

ARN: arn:aws:logs:us-east-1:095074107161:log-group:/aws/lambda/PaymentLambda:

Creation time: 5 minutes ago

Retention: Never expire

Stored bytes: -

Metric filters: 0

Subscription filters: 0

Contributor Insights rules: -

KMS key ID: -

Deletion protection: Off

Data protection: -

Sensitive data count: -

Custom field indexes: Configure

Transformer: Configure

Anomaly detection: Configure

Log streams (1)

By default, we only load the most recent log streams.

Filter log streams or try prefix search

Exact match

Show expired

Info

Log stream	Last event time
2025/12/11/[\$LATEST]43d0809ced9d4e78a3a787d02d65b050	2025-12-11 06:09:44 (UTC)